STEMI — Stents, Restenosis & Stent Thrombosis

1. 12-month calendar / DAPT duration

WHAT YOU SEE

Calendar reading 12 MONTHS on the upper shelf.

WHAT IT ENCODES

After PCI with a drug-eluting stent, default DAPT (aspirin + P2Y12 inhibitor) is 12 months for ACS. High-bleeding-risk patients may shorten to 1-6 months; very high ischemic risk may extend beyond 12. Stable CAD elective PCI needs only ~6 months.

BOARD TRAP

Don't quote 1 month as standard — that short course is only for high-bleeding-risk patients, not routine ACS, where 12 months is the answer.

2. 'Stroke / brain first — coronary anatomy first' sign

WHAT YOU SEE

Placard reading 'stroke-brain' with 'CORONARY ANATOMY FIRST' on the fence.

WHAT IT ENCODES

Define the coronary anatomy with angiography before committing to stent type or surgery. Complex multivessel/left-main disease may favor CABG over PCI; SYNTAX score guides the decision. Anatomy drives revascularization strategy.

BOARD TRAP

Stenting every lesion is wrong in high SYNTAX-score multivessel or left-main disease — CABG gives better survival there.

3. BMS jar — 20-50% restenosis

WHAT YOU SEE

First glass jar labeled bare-metal-stent with '20-50% RESTENOSIS'.

WHAT IT ENCODES

Bare-metal stents (BMS) have the highest in-stent restenosis (20-50%) due to neointimal hyperplasia. They require shorter DAPT but restenose far more than DES. Largely replaced by DES.

BOARD TRAP

Restenosis (gradual neointimal regrowth, months later, presents as recurrent angina) is NOT stent thrombosis (abrupt thrombotic occlusion presenting as acute MI).

4. DES jar — 5-15% restenosis

WHAT YOU SEE

Drug-eluting-stent jar marked '5-15%'.

WHAT IT ENCODES

Drug-eluting stents elute antiproliferative drugs (sirolimus/everolimus) that suppress neointimal hyperplasia, cutting restenosis to 5-15%. Newer second-generation DES have the lowest event rates.

BOARD TRAP

DES reduce restenosis but historically raised LATE thrombosis risk — hence the longer DAPT requirement vs BMS.

5. Clock — 'more late thrombosis'

WHAT YOU SEE

Clock with 'MORE LATE THROMBOSIS' beside the DES jar.

WHAT IT ENCODES

First-generation DES delayed endothelialization, raising late (>30 day) and very-late (>1 yr) stent thrombosis risk. This is why DES historically mandated prolonged DAPT. Second-generation DES largely fixed this.

BOARD TRAP

Late stent thrombosis is the classic consequence of premature DAPT discontinuation — stopping the P2Y12 inhibitor early is the most common precipitant.

6. 'NOT DAPT' crossed pill bottle

WHAT YOU SEE

Crossed-out pill bottle labeled 'NOT DAPT' on the THROMBOSIS bench.

WHAT IT ENCODES

Inadequate or interrupted dual antiplatelet therapy is the leading modifiable cause of stent thrombosis. Premature cessation (e.g., for surgery) carries the highest risk; bridge or defer elective surgery when possible.

BOARD TRAP

Subtherapeutic single antiplatelet is not enough after stenting — both agents are needed for the recommended duration to prevent thrombosis.

7. Knight pierced by sword — THROMBOSIS

WHAT YOU SEE

Armored knight skewered on the bench labeled THROMBOSIS.

WHAT IT ENCODES

Stent thrombosis is an abrupt thrombotic occlusion of the stent → acute STEMI in the stented territory, high mortality. Timing: acute (<24h), subacute (1-30d), late (>30d), very late (>1yr).

BOARD TRAP

Sudden re-presentation with STEMI weeks after PCI = stent thrombosis, not restenosis (which is gradual exertional angina).

8. Horse 'LOAD' — loading dose

WHAT YOU SEE

Horses on a shelf labeled LOAD.

WHAT IT ENCODES

Antiplatelet loading doses are given at PCI: aspirin 325 mg plus a P2Y12 loading dose (clopidogrel 600 mg, ticagrelor 180 mg, or prasugrel 60 mg). Prasugrel/ticagrelor are preferred over clopidogrel in ACS.

BOARD TRAP

Prasugrel is contraindicated with prior stroke/TIA and avoided in age >75 or weight <60 kg — don't reflexively load it.

9. '3-9x' fallen knight — risk multiplier

WHAT YOU SEE

Collapsed knight with a giant '3-9x' burst.

WHAT IT ENCODES

Stopping DAPT prematurely multiplies stent thrombosis risk several-fold. The thrombotic event rate after early discontinuation is dramatically higher, which is why timing of any interruption is a board favorite.

BOARD TRAP

Elective non-cardiac surgery should be deferred (≥6 months after DES if possible); if urgent, continue aspirin and minimize the gap.

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10. 'DEATH 10-20%' / 'MI 50-70%' tags

WHAT YOU SEE

Tags reading DEATH 10-20% and MI 50-70%.

WHAT IT ENCODES

Stent thrombosis is catastrophic: roughly 10-20% mortality and 50-70% present as MI. This lethality justifies aggressive antiplatelet adherence and careful procedural technique (adequate stent expansion).

BOARD TRAP

These grim numbers belong to stent THROMBOSIS, not restenosis — restenosis is usually a benign recurrent-angina problem, rarely fatal.

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11. Crossed knights — RESTENOSIS

WHAT YOU SEE

Two knights with a big red X over a RESTENOSIS sign.

WHAT IT ENCODES

In-stent restenosis is neointimal hyperplasia narrowing the stent over months, presenting as recurrent stable/progressive angina. Risk factors: diabetes, small vessels, long stents, BMS over DES.

BOARD TRAP

Restenosis is gradual and presents as recurrent angina — not the abrupt MI of stent thrombosis. DES, not BMS, minimizes it.

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12. Anvil 'NEOINTIMAL HYPERPLASIA'

WHAT YOU SEE

Anvil and forge labeled neointimal hyperplasia, '12 MONTHS — GRADUAL'.

WHAT IT ENCODES

Restenosis mechanism = smooth-muscle proliferation/neointimal hyperplasia, evolving gradually over ~12 months. DES antiproliferative coating directly targets this process. Treat with repeat PCI (DES or drug-coated balloon).

BOARD TRAP

Antiproliferative DES drugs reduce neointimal hyperplasia/restenosis but slow endothelial healing — the trade-off that historically increased late thrombosis.
